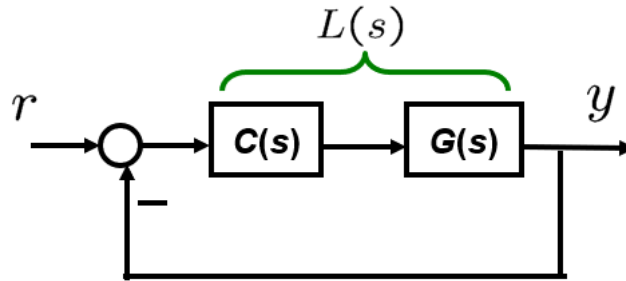


Assignment 20 (ELEC 341 L20_FreqDomSpec)

Problem 1:

Given the unity feedback system below, use frequency response methods to determine the value of gain, $C(s)$, to yield a step response with a 20% overshoot. Graph the step response.



$$G(s) = \frac{(s+4)}{s(s+8)(s+10)(s+15)}$$

$$C(s) = K.$$

$$1 + K G(s) = 0.$$

$$K = \frac{1}{|G(j\omega)|}$$

$$0.605 = 100e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}, \quad \angle G(j\omega) = -180^\circ + PM_{comp}.$$

$$\zeta = \frac{-\ln\left(\frac{0.605}{100}\right)}{\sqrt{\pi^2 + \ln^2\left(\frac{0.605}{100}\right)}}, \quad PM_{comp} = 100\%$$

$$\zeta = 0.455949$$

$$PM_{comp} = 45.59498^\circ.$$

$$\begin{aligned} &= s(s+8)(s^2+25s+250) \\ &= s(s^2+25s+250s+8s^2+200s+2000) \\ &= s^4+33s^3+450s^2+2000s \end{aligned}$$

$$G(j\omega) = \frac{j\omega+4}{(j\omega)(j\omega+8)(j\omega+10)(j\omega+15)}, \quad |G(j\omega)| = \frac{\sqrt{\omega^2+16}}{\omega\sqrt{\omega^2+64}\sqrt{\omega^2+100}\sqrt{\omega^2+225}}$$

$$\angle G(j\omega) = \tan^{-1}\left(\frac{\omega}{4}\right) - 90^\circ - \tan^{-1}\left(\frac{\omega}{8}\right) - \tan^{-1}\left(\frac{\omega}{10}\right) - \tan^{-1}\left(\frac{\omega}{15}\right)$$

$$\angle G(j\omega_g) = -180^\circ + PM_{comp}$$

$$\angle G(j\omega_g) = -134.4050189^\circ$$

$$\angle G(j\omega_g) + 134.4050189^\circ = 0.$$

$$\omega_g = 7.4654 \text{ rad/s.}$$

$$|G(j\omega_g)| = 4.9586469 \times 10^{-4}$$

$$C(s) = K = \frac{1}{|G(j\omega_g)|}$$

$$C(s) = 2016.679156$$

